

Script Inputs and Outputs

This file contains the quantitative analysis pipeline for the IADB audit study. The scripts should be run in numerical order.

Unless otherwise noted, paths are written relative to the project root.

01_data_cleaning_v2.R

Purpose

Cleans the raw SurveyCTO export, constructs treatment variables, outcome variables, KYC measures, time variables, cost variables in local currency, and quality flags.

Required inputs

- IADB_Survey_WIDE_may16.csv

Main outputs

- data/clean/IADB_surveycto_clean_may16.csv
- data/clean/IADB_surveycto_clean_may16.rds
- data/clean/IADB_surveycto_clean_diagnostics_may16.rds
- data/clean/IADB_surveycto_balance_by_channel_may16.csv

Optional/commented outputs

These are present in the script but currently commented out:

- data/clean/IADB_surveycto_missing_by_channel_may16.csv
 - data/clean/IADB_surveycto_quality_flags_may16.csv
 - data/clean/IADB_surveycto_kyc_validation_may16.csv
-

02_enriched_payment_schedule.R

Purpose

Loads the randomized transaction schedules and the internal payment-tracking sheet, harmonizes names/IDs/channels/dates, and creates an enriched payment schedule that links planned randomized slots to actual payment information.

Required inputs

- `data/raw/[IADB] - Internal Payment Tracking - Payment Schedule.csv`
- `IADB/data/randomization/master_schedule_feb13.csv`
- `IADB/data/randomization/master_schedule_feb26.csv`
- `IADB/data/randomization/master_schedule_mar12.csv`
- `IADB/data/randomization/master_schedule_mar30_may15.csv`
- `IADB/data/randomization/master_schedule_apr24_may25.csv`

Main outputs

- `data/clean/sap_dataset_builder/IADB_payment_schedule_enriched_with_randomization.csv`
- `data/clean/sap_dataset_builder/IADB_payment_schedule_enriched_with_randomization.rds`

Diagnostic outputs

- `data/clean/sap_dataset_builder/IADB_enriched_schedule_checks.csv`
- `data/clean/sap_dataset_builder/IADB_enriched_schedule_by_status.csv`
- `data/clean/sap_dataset_builder/IADB_enriched_schedule_duplicate_slots.csv`
- `data/clean/sap_dataset_builder/IADB_enriched_schedule_review_needed.csv`

Optional/commented outputs

These are present in the script but currently commented out:

- `data/clean/sap_dataset_builder/IADB_exact_match_checks.csv`
- `data/clean/sap_dataset_builder/IADB_exact_match_duplicate_slots.csv`

03_build_sap_dataset.R

Purpose

Builds the core SAP dataset by matching cleaned SurveyCTO records to randomized/payment schedule slots. This script creates the schedule-level denominator, the observed-attempt sample, the per-protocol sample, and key diagnostics for unmatched rows, duplicates, protocol deviations, and manual review.

Required inputs

- `data/clean/IADB_surveycto_clean_may16.csv`
- `data/clean/sap_dataset_builder/IADB_payment_schedule_enriched_with_randomization.csv`

Conditional/manual inputs

- `data/manual/IADB_confederate_crosswalk.csv`

Used only if SurveyCTO confederate IDs and schedule confederate names require manual harmonization.

- `data/clean/sap_dataset_builder/IADB_manual_schedule_match_review_completed.csv`

Required if `SKIP_MANUAL_REVIEW <- FALSE`. If this file does not exist, the script exports a template for manual review and stops.

Main outputs

- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base.csv`
- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.csv`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.rds`
- `data/clean/sap_dataset_builder/IADB_sap_attempted_after_funding.csv`
- `data/clean/sap_dataset_builder/IADB_sap_attempted_after_funding.rds`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol.csv`

- `data/clean/sap_dataset_builder/IADB_sap_per_protocol.rds`

Matching/audit outputs

- `data/clean/sap_dataset_builder/IADB_manual_schedule_match_review_to_complete.csv`
- `data/clean/sap_dataset_builder/IADB_03_schedule_checks.csv`
- `data/clean/sap_dataset_builder/IADB_03_best_schedule_matches.csv`
- `data/clean/sap_dataset_builder/IADB_03_sap_merge_checks.csv`
- `data/clean/sap_dataset_builder/IADB_03_surveycto_schedule_matched_full_audit.csv`
- `data/clean/sap_dataset_builder/IADB_03_surveycto_schedule_matched_dedup.csv`
- `data/clean/sap_dataset_builder/IADB_03_surveycto_schedule_matched_dedup.rds`

Diagnostic/review outputs

- `data/clean/sap_dataset_builder/IADB_03_incomplete_manual_review_rows.csv`
- `data/clean/sap_dataset_builder/IADB_03_bad_manual_slot_ids.csv`
- `data/clean/sap_dataset_builder/IADB_03_final_duplicate_schedule_slots.csv`
- `data/clean/sap_dataset_builder/IADB_03_survey_rows_excluded_from_sap_firstpass.csv`
- `data/clean/sap_dataset_builder/IADB_03_unmatched_survey_rows.csv`
- `data/clean/sap_dataset_builder/IADB_03_non_attempted_schedule_slots.csv`
- `data/clean/sap_dataset_builder/IADB_03_protocol_deviations.csv`
- `data/clean/sap_dataset_builder/IADB_03_bad_survey_completion_review.csv`

04_build_analysis_sample_and_attrition.R

Purpose

Builds the maximal automated matched sample, preserves the conservative first-pass sample, and creates sample-comparison and quality diagnostics. This script is the bridge between the initial matched SAP dataset and the final main/sensitivity samples used in the SAP models.

Required inputs

- `data/clean/sap_dataset_builder/IADB_03_surveycto_schedule_matched_full_audit.csv`
- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.rds`

Main outputs

- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base_maximal_auto.csv`
- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.csv`
- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto.csv`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto.rds`

Diagnostic outputs

- `data/clean/sap_dataset_builder/IADB_sap_maximal_auto_needs_manual_review.csv`
- `data/clean/sap_dataset_builder/IADB_04_sample_comparison.csv`
- `data/clean/sap_dataset_builder/IADB_04_sample_quality_summary.csv`

Optional/commented outputs

- `data/clean/sap_dataset_builder/IADB_04_slot_conflict_diagnostics.csv`

04b_duplicate_slot_recovery_audit.R

Purpose

Audits duplicate schedule-slot matches and identifies possible recovery candidates among SurveyCTO rows that may have been excluded because of duplicate-slot conflicts.

This script is diagnostic. It does not directly change the main analysis sample.

Required inputs

- `data/clean/sap_dataset_builder/IADB_03_surveycto_schedule_matched_full_audit.csv`
- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.rds`

Outputs

- `data/clean/sap_dataset_builder/IADB_04b_duplicate_slot_diagnostics.csv`
 - `data/clean/sap_dataset_builder/IADB_04b_duplicate_rows_ranked.csv`
 - `data/clean/sap_dataset_builder/IADB_04b_nonselected_duplicate_rows.csv`
 - `data/clean/sap_dataset_builder/IADB_04b_top_recovery_candidates.csv`
 - `data/clean/sap_dataset_builder/IADB_04b_recovery_summary.csv`
-

04c_prepare_recovery_review.R

Purpose

Prepares a manual review file for strong recovery candidates identified by the duplicate-slot recovery audit. This script helps decide whether excluded SurveyCTO rows should be reassigned to unused schedule slots.

This script is diagnostic/manual-review oriented. It does not automatically change the main analysis sample unless its decisions are later incorporated.

Required inputs

The script looks for either of the following:

- `data/clean/sap_dataset_builder/IADB_04a_top_recovery_candidates.csv`
- `data/clean/sap_dataset_builder/IADB_04b_top_recovery_candidates.csv`

In the current sequence, the expected input is:

- `data/clean/sap_dataset_builder/IADB_04b_top_recovery_candidates.csv`

Outputs

- `data/clean/sap_dataset_builder/IADB_04c_all_strong_recovery_candidates.csv`
- `data/clean/sap_dataset_builder/IADB_04c_candidate_slot_pressure.csv`
- `data/clean/sap_dataset_builder/IADB_04c_strong_recovery_candidates_for_review.csv`
- `data/clean/sap_dataset_builder/IADB_04c_recovery_decisions_template.csv`
- `data/clean/sap_dataset_builder/IADB_04c_recovery_review_summary.csv`

05_run_sap_models.R

Purpose

Runs the core SAP models for transaction success and KYC burden. It estimates the main model on the maximal automated sample and sensitivity models using clean, per-protocol, and conservative samples.

Required inputs

- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.rds`

Outputs

Saved in:

- `data/clean/sap_dataset_builder/sap_results_maximal_auto/`

Specific outputs:

- `sap_sample_summary.csv`
 - `sap_main_by_channel.csv`
 - `sap_models_skipped.csv`
 - `sap_success_kyc_main_and_sensitivity.txt`
 - `sap_success_kyc_main_and_sensitivity.tex`
 - `sap_success_kyc_main_and_sensitivity.rds`
-

06_exchange_rates_fred.R

Purpose

Builds daily FX rates needed to convert local-currency cost variables into USD. The output is used by the cost/time model script.

Required inputs

The script can run using its internal currency/date logic, but it uses existing SAP datasets to infer the required date range when available.

Candidate SAP inputs:

- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.rds`

External/API inputs

- FRED API access, when available, for supported currencies.
- Public no-key currency API fallback for currencies not covered by FRED or when FRED is unavailable.

Outputs

Saved in:

- `data/manual/`

Specific outputs:

- `data/manual/IADB_fx_rates_daily.csv`

- `data/manual/IADB_fx_rates_daily_diagnostics.csv`
 - `data/manual/IADB_fx_rates_daily_source_counts.csv`
-

07_cost_time_sap_models.R

Purpose

Adds FX conversion and time-cost measures, creates cost/time analysis samples, and runs SAP models for monetary cost, settlement time, reported transaction duration, and interaction time.

Required inputs

- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.rds`
- `data/manual/IADB_fx_rates_daily.csv`

Optional inputs

- `data/manual/IADB_hourly_wage_lookup.csv`

Used if available to construct additional time-cost measures.

Intermediate dataset outputs

Saved in:

- `data/clean/sap_dataset_builder/`

Specific outputs:

- `IADB_sap_observed_maximal_auto_cost_time.csv`
- `IADB_sap_observed_maximal_auto_cost_time.rds`
- `IADB_sap_observed_clean_cost_time.csv`
- `IADB_sap_observed_clean_cost_time.rds`
- `IADB_sap_per_protocol_maximal_auto_cost_time.csv`
- `IADB_sap_per_protocol_maximal_auto_cost_time.rds`
- `IADB_sap_observed_conservative_cost_time.csv`

- `IADB_sap_observed_conservative_cost_time.rds`

Diagnostic outputs

Saved in:

- `data/clean/sap_dataset_builder/sap_results_cost_fx_time/`

Specific outputs:

- `IADB_06_country_currency_lookup.csv`
- `IADB_06_wage_lookup_used.csv`
- `IADB_06_fx_daily_completed.csv`
- `IADB_06_fx_coverage_summary.csv`
- `IADB_06_fx_missing_review.csv`
- `IADB_06_cost_time_sample_summary.csv`
- `IADB_06_cost_time_outlier_review.csv`
- `IADB_06_cost_time_by_channel_main.csv`
- `IADB_06_fx_sources_used_main_cost_rows.csv`
- `IADB_06_model_sample_summary.csv`
- `IADB_06_cost_time_models_skipped.csv`

Model outputs

Saved in:

- `data/clean/sap_dataset_builder/sap_results_cost_fx_time/`

Specific outputs:

- `IADB_06_cost_time_models.rds`
- `IADB_06_cost_any_attempt_models.txt`
- `IADB_06_cost_any_attempt_models.tex`
- `IADB_06_cost_success_only_models.txt`
- `IADB_06_cost_success_only_models.tex`
- `IADB_06_reported_time_models.txt`
- `IADB_06_reported_time_models.tex`
- `IADB_06_interaction_time_models.txt`
- `IADB_06_interaction_time_models.tex`
- `IADB_06_transaction_duration_models.txt`
- `IADB_06_transaction_duration_models.tex`

- `IADB_06_cost_time_models_all.txt`
- `IADB_06_cost_time_models_all.tex`

Note on time variables

In the current pipeline:

- `time_hours` is the PAP-aligned settlement-time variable.
- `transaction_duration_hours` is based on `i1_exact_minutes / 60`.
- `interaction_time_hours` is travel time + waiting time + service time, divided by 60.

Downstream labels should distinguish these concepts clearly.

`08_generate_final_results.R`

Purpose

Generates the final `fixest::etable` outputs across all main outcome families: success, KYC, cost, settlement time, interaction time, and reported transaction duration. Those tables are for internal purposes only.

This script does not clean, merge, or transform raw data. It loads already-prepared analysis datasets and estimates/export final model tables.

Required inputs

Success/KYC datasets:

- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_first_pass.rds`

Cost/time datasets:

- `data/clean/sap_dataset_builder/IADB_sap_observed_maximal_auto_cost_time.rds`
- `data/clean/sap_dataset_builder/IADB_sap_observed_clean_cost_time.rds`

- data/clean/sap_dataset_builder/IADB_sap_per_protocol_maximal_auto_cost_time.rds
- data/clean/sap_dataset_builder/IADB_sap_observed_conservative_cost_time.rds

Outputs

Saved in:

- data/clean/sap_dataset_builder/final_etables/

Model/sample diagnostics:

- IADB_07_model_sample_summary.csv
- IADB_07_skipped_models.csv
- IADB_07_all_models_full_list.rds
- IADB_07_all_models_nonnull.rds

Final etable outputs:

- IADB_07_main_outcomes_models.txt
- IADB_07_main_outcomes_models.tex
- IADB_07_main_outcomes_models.rds
- IADB_07_success_models.txt
- IADB_07_success_models.tex
- IADB_07_success_models.rds
- IADB_07_kyc_models.txt
- IADB_07_kyc_models.tex
- IADB_07_kyc_models.rds
- IADB_07_cost_any_attempt_models.txt
- IADB_07_cost_any_attempt_models.tex
- IADB_07_cost_any_attempt_models.rds
- IADB_07_cost_success_only_models.txt
- IADB_07_cost_success_only_models.tex
- IADB_07_cost_success_only_models.rds
- IADB_07_reported_time_models.txt
- IADB_07_reported_time_models.tex
- IADB_07_reported_time_models.rds
- IADB_07_interaction_time_models.txt
- IADB_07_interaction_time_models.tex

- `IADB_07_interaction_time_models.rds`
 - `IADB_07_transaction_duration_models.txt`
 - `IADB_07_transaction_duration_models.tex`
 - `IADB_07_transaction_duration_models.rds`
 - `IADB_07_all_models.txt`
 - `IADB_07_all_models.tex`
 - `IADB_07_all_models.rds`
-

09_attrition.R

Purpose

Runs implementation and attrition checks using the schedule-level denominator. The goal is to test whether assigned treatment arms predict being attempted, skipped, funded but not attempted, successfully completed, or implemented with protocol deviations.

Required inputs

The script reads the first available file from this priority list:

- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base_maximal_auto.rds`
- `data/clean/sap_dataset_builder/IADB_sap_schedule_level_base.rds`

Outputs

Saved in:

- `data/clean/sap_dataset_builder/implementation_checks/`

Dataset outputs:

- `IADB_09_schedule_level_column_inventory.csv`
- `IADB_09_implementation_base.rds`
- `IADB_09_implementation_base.csv`

Summary outputs:

- `IADB_09_overall_implementation_summary.csv`
- `IADB_09_implementation_by_assigned_arm.csv`

- `IADB_09_implementation_by_confederate.csv`

Model outputs:

- `IADB_09_implementation_attrition_models.rds`
 - `IADB_09_implementation_attrition_models.txt`
 - `IADB_09_implementation_attrition_models.tex`
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Pipeline Dependency Summary

The main dependency chain is:

1. `01_data_cleaning_v2.R` produces cleaned SurveyCTO data.
2. `02_enriched_payment_schedule.R` produces enriched randomized/payment schedule.
3. `03_build_sap_dataset.R` merges cleaned SurveyCTO data with enriched schedule and creates first-pass SAP samples.
4. `04_build_analysis_sample_and_attrition.R` creates maximal-auto, clean, per-protocol, and conservative analysis samples.
5. `04b_duplicate_slot_recovery_audit.R` and `04c_prepare_recovery_review.R` provide optional duplicate/recovery diagnostics.
6. `05_run_sap_models.R` runs success and KYC models.
7. `06_exchange_rates_fred.R` builds FX-rate inputs for USD cost conversion.
8. `07_cost_time_sap_models.R` creates cost/time datasets and runs cost/time models.
9. `08_generate_final_results.R` produces final model tables.
10. `09_attrition.R` produces implementation and attrition diagnostics using the schedule-level denominator.